



UNIVERSIDADE D
COIMBRA

Knowledge and Language 2025/2026 - Project Statement

*Masters in Informatics Engineering
Natural Language Processing and Knowledge
Representation*

Membros:

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1 Problem Statement

General LLMs can answer many questions but often lack domain or culturally specific Knowledge. The challenge we aim to address is building a system that can reliably and accurately answer complex, domain-specific questions about Portuguese cuisine and other recipes using structured data from the food.com dataset.

2 Project Goal

The goal of the project is to develop a ChatBot that combines a structured Knowledge Graph (KG) of recipes with Natural Language Processing (NLP) techniques and Retrieval-Augmented Generation (RAG). The ChatBot will be able to interpret user queries in natural language, retrieve the most relevant recipes or recipe attributes, and provide accurate, context-aware responses.

3 Data and Tools and Proposed Approach

We will treat the dataset to include more of Portuguese cuisine recipes and keep the distribution of other cuisines. The structured Knowledge Graph (KG) is built using the RDF (Resource Description Framework) framework. Each RDF triplet has the form:

- **subject:** a specific recipe or ingredient (e.g., "Bacalhau à Brás"),
- **predicate:** an attribute such as `hasIngredient` or `hasPrepTime`,
- **object:** the value of the attribute (e.g., "tomato", "15 minutes").

The system first uses a fine-tuned BERT [1] model to identify the user's intent, such as finding a recipe and retrieving ingredients. Next, spaCy extracts relevant entities from the query, including ingredients, name of the recipe and cooking time, which are then linked to corresponding nodes or properties in the Knowledge Graph. The intent and linked entities are combined to generate a SPARQL query for precise retrieval of structured information. Each query has more context from the dataset, such as description, cooking steps and other tags (with information like diet, food restrictions etc) added with a RAG step to give Ollama more context through an embeddings-based retrieval that identifies the top-k most relevant recipes from textual fields, and a Retrieval-Augmented Generation (RAG). Finally, a React-based front-end presents the answers to the user in a conversational interface.

4 checkpoint dates

Checkpoint 1 – > 18 November 2025 Checkpoint 2 – > 2 December 2025

References

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” *arXiv preprint arXiv:1810.04805*, 2019, submitted 11 Oct 2018 (v1); last revised 24 May 2019 (v2). [Online]. Available: <https://doi.org/10.48550/arXiv.1810.04805>